

DEEP PROCESSING OF PLANT PROTEIN

Status: Open Source Patent for Public Use

Author: Grandfather Savely's Universe

License: Heritage of humanity. No corporation may monopolize this algorithm. Free use is encouraged with attribution to the "Grandfather Savely's Universe" methodology.

1. Introduction: Rehabilitating Soy and the "Hacking" of Biological Defense

Intellect as the Primary Digestive Organ

In the wild, every animal is confined to its narrow biological niche. The koala neutralizes eucalyptus toxins. The panda digests tough bamboo. Their organisms have been "sharpened" for millions of years against the specific antinutrients of a single plant.

Humans are the only beings whose digestive system extends beyond the body and is governed by intellect. We don't need to wait for our gastrointestinal tract to evolve in order to adapt to the defense toxins of soy or corn. We have a brain to deactivate these poisons while still in the pot.

We survived as a species not because our stomachs became stronger, but because our minds learned to make any food wholesome — hacking nature's locks where other animals would simply perish.

Why Soy?

Soy is the most efficient, cheapest, and most complete source of plant protein on the planet. But two barriers stand in the way of its assimilation:

- the plant's natural defenses (antinutrients)
- modern agricultural methods (chemicals)

For decades, the world has been divided into two camps: one praises soy for its amino acid profile, while the other — justifiably — criticizes it for poor digestibility, "heaviness" on the GI tract, and the presence of lectins.

We assert: the problem is not soy itself, but the absence of a culture of deep preparation.

What This Method Provides

The "hacking" technology transforms soy into a product purer than Western Bio and Organic standards. While "organic" labels guarantee only soil purity, our method guarantees the purity of the product's cell itself — complete cleansing of the plant's natural chemical weaponry and residual agrochemical traces.

The result is a 100% biological concentrate, equally ideal for vegetarians and professional athletes.

Why It Works

Any grain is a "safe" that nature has locked with a chemical combination to prevent it from being eaten before it can sprout. We do not crack the safe with a "crowbar" (high temperature) that destroys the contents. We use gentle biochemical decryption:

1. Deactivation of "chemical weapons" — neutralization of enzyme inhibitors that prevent the body from digesting protein.
2. Removal of "armor" — complete elimination of the hull, the main carrier of antinutrients and bitter taste.
3. Cleansing of agro-residues — the alkaline environment and gentle temperature regime "wash" the product of residual fertilizers and protective coatings.

2. Historical Precedent: Parallel with Nixtamalization

Our method is not an artificial invention, but a restoration of lost engineering knowledge that humanity used for millennia.

The clearest example is nixtamalization of corn, created by the civilizations of Mesoamerica. Ancient peoples understood: corn boiled in plain water leads to illness and exhaustion (pellagra). But when cooked in an alkaline medium (with lime or ash),

"magic" occurs — bound vitamin B₃ (niacin) and amino acids, previously locked inside the grain, are released.

The alkaline medium turns "empty" food into a superfood. Our method does the same for soy at 65°C in a mildly alkaline solution. We are not simply heating beans — we are breaking the chemical bonds that hold vitamins and minerals captive to antinutrients.

Our ancestors intuitively observed this principle through hours of slow simmering of grains and vegetables at a gradually falling temperature. This allowed the grain's enzymes to work for us, not against us.

Conclusion: Just as corn without alkali is poison, soy without proper "hacking" is ballast. We return the product to its true biological value.

This document is a bridge between ancestral traditions, who intuitively understood the value of slow cooking, and modern systemic approaches to food safety.

3. Technological Cycle

Stage 1. Hydrothermal Hacking and Dehulling (Cleaning)

During soaking, the hull swells faster than the kernel. While the soy is still in water, light mechanical action (rubbing) causes the hull to detach. Since it is lighter than the kernel, it floats (flotation).

Result: a clean, "naked" kernel, stripped of 90% of natural defense toxins. Without this stage, the protein remains "dirty," with a grayish tint and excess biological load.

Stage 2. Alkaline Hydrothermal Treatment

Parameters: heating to 65°C for 12 hours in a mildly alkaline solution (potassium carbonate or baking soda).

- Why 65°C? This is the deactivation point for enzyme inhibitors (trypsin). We "disable" the grain's biological defense without denaturing the protein itself.
- Why alkali? It softens cell walls, helps dissolve hard-to-reach antinutrients, and neutralizes the specific legume smell (lipoxygenase).

Stage 3. Dispersion (Grinding)

Choice depends on scale:

Option	Equipment	Result
Industrial	Colloid mill	Nano-fraction, "silky" protein, ideal for freeze-drying
Craft/home	Blender	Coarser fraction, texture resembling meat fibers or dense cottage cheese

Stage 4. Directed Coagulation (Curdling While Preserving Life)

Agents:

- citric acid (natural acidifier)
- food-grade gypsum (calcium sulfate) — a gentle structuring agent
- starter culture (kefir or kimchi cabbage) — source of living enzymes

Process:

Heating the suspension to 65°C in the presence of citric acid and gypsum.

What happens:

1. The protein begins to form a matrix
2. Gypsum gently fixes the structure without killing enzymes
3. Soy enzymes and starter culture enzymes remain active

Why this is important:

At 65°C, we are at the upper boundary where:

- antinutrients are already deactivated
- the protein is already coagulating
- enzymes are still alive

If heated higher — enzymes die. If gypsum is not added — the structure remains loose and unstable.

The Golden Mean of the Technology:

65°C + citric acid + gypsum + living starter culture = coagulation without killing the enzymatic system.

In this case, the following are fully preserved:

- native soy enzymes
- fermentation enzymes (kefir starter or kimchi cabbage starter)

Result: we obtain not a dead protein, but a living, enzymatically active protein matrix that continues to work on assimilation once inside the body.

Stage 5. Hydration and Drying

For business / military (freeze-drying):

Excess moisture is removed from the protein mass, and the product is dried while preserving micropores. Thanks to the pores, the dry concentrate instantly absorbs water during preparation — complete food in 2 minutes.

For home / field conditions:

Sun drying or shade drying (as in kurut preparation). The technology works without expensive equipment.

4. Final Product Characteristics

Parameter	Value
Amino acid profile	Complete, closely approaching reference animal protein
Bioavailability	95–98% (due to hull removal, inhibitor deactivation, and enzymatic activity)

Environmental barrier	The method reduces the agrochemical footprint through alkaline washing and bacterial sorption
Structure	Protein + grain fiber (not an isolated protein, but a complex ideally suited to human digestion)
Enzymatic activity	Preserved (native soy enzymes + starter culture enzymes)

5. Adaptation for Different Conditions

Home Conditions ("Kitchen Chemistry")

The technology is fully adapted. At home, even deeper cleansing is possible using bacterial starter cultures.

We are working on creating:

- a detailed step-by-step recipe
- video instructions

Additional materials will be posted as they become available.

Industrial Scale

For large producers, the method presents no difficulty. Cleaned soy (without hull and skin) is already 50% of the finished product.

6. Legal Status

This technology is the heritage of humanity. No corporation may monopolize this algorithm.

Free use is encouraged with attribution to the "Grandfather Savely's Universe" methodology.

Document published May 3, 2026. Russian and English versions. Status: Open Source Patent for Public Use.